Trace-small imaginary parts preserve spectral symbols

Principia Math in Progress

September 12, 2026

Task 1. Solve the real-part spectral-symbol conjecture

We prove Conjecture 2 on page 6 of Barbarino's *Conjectures on Perturbations of Hermitian Sequences* [1], corresponding to MathDB 340128. For a matrix X, put $\operatorname{Re} X = (X + X^*)/2$ and $\operatorname{Im} X = (X - X^*)/(2i)$; $\|X\|_1$ denotes the sum of its singular values. For $n \times n$ matrices, write $X_n \sim_\lambda f$ when

$$\frac{1}{n}\sum_{j=1}^{n} F(\lambda_j(X_n)) \longrightarrow \frac{1}{|D|}\int_D F(f(x))\,dx \qquad (F \in C_c(\mathbb{C})), \tag{1}$$

where $f : D \to \mathbb{C}$ is measurable and $0 < |D| < \infty$. Eigenvalues are counted with algebraic multiplicity.

Theorem 1. *If $X_n \sim_\lambda f$ and $\|\operatorname{Im} X_n\|_1 = o(n)$, then $\operatorname{Re} X_n \sim_\lambda f$. In particular f is real almost everywhere. No bound on $\|X_n\|$ is required.*

The main estimate is a finite-dimensional weak trace-norm inequality for triangular truncation. Such inequalities are classical; see, for example, the estimate cited in the proof of Lemma 3.2 in [2]. We include an elementary determinant proof, so the argument does not depend on that result.

For Hermitian $B = (b_{jk})$, let $U(B)$ be its strictly upper triangular part and define the Hermitian matrix

$$\mathcal{H}(B) = i\big(U(B) - U(B)^*\big).$$

For any matrix A, let $N_t(A) = \#\{j : s_j(A) > t\}$, for $t > 0$.

Lemma 1. *For every Hermitian matrix B of any size,*

$$N_t(\mathcal{H}(B)) \le \frac{4}{\log 2}\frac{\|B\|_1}{t}. \tag{2}$$

Proof. First suppose $B \ge 0$ and write $H = \mathcal{H}(B)$. The matrix $B + iH = \operatorname{diag}(B) + 2U(B)^*$ is lower triangular. Hence

$$|\det(I + B/t + iH/t)| = \prod_j (1 + b_{jj}/t) \le \exp(\operatorname{tr} B/t). \tag{3}$$

We also have

$$|\det(I + B/t + iH/t)| \ge |\det(I + iH/t)|. \tag{4}$$

Indeed, put $P = B/t \ge 0$, $Q = H/t = Q^*$, and $Z = P^{1/2}(I + iQ)^{-1}P^{1/2}$. Since

$$\operatorname{Re} Z = P^{1/2}(I + Q^2)^{-1}P^{1/2} \ge 0,$$

every eigenvalue z of Z satisfies $\operatorname{Re} z \geq 0$. Thus $|\det(I+Z)| = \prod_z |1+z| \geq 1$. Factoring $I+P+iQ$ and using $\det(I+AB) = \det(I+BA)$ proves (4).

Combining (3)–(4) and taking logarithms gives

$$\frac{1}{2}\sum_j \log\bigl(1+\lambda_j(H)^2/t^2\bigr) \leq \frac{\operatorname{tr} B}{t}.$$

Each term with $|\lambda_j(H)| > t$ exceeds $(\log 2)/2$. Consequently $N_t(H) \leq 2\operatorname{tr} B/(t\log 2)$.

For general B, use its positive and negative spectral parts: $B = B_+ - B_-$, with $B_\pm \geq 0$ and $\operatorname{tr} B_+ + \operatorname{tr} B_- = \|B\|_1$. Linearity and the singular-value counting inequality give

$$N_t(\mathcal{H}(B)) \leq N_{t/2}(\mathcal{H}(B_+)) + N_{t/2}(\mathcal{H}(B_-)) \leq \frac{4\|B\|_1}{t\log 2}.$$

For completeness, the counting inequality follows by splitting each summand into its singular directions above $t/2$ and a remainder of norm at most $t/2$. The sum of the former pieces has rank at most the sum of the two counts, and the total remainder has norm at most t. □

Lemma 2. *Suppose A_n, D_n are Hermitian, of size n, and $N_t(A_n - D_n) = o(n)$ for every fixed $t > 0$. Then, for every $g \in C_c(\mathbb{R})$,*

$$\frac{1}{n}\operatorname{tr} g(A_n) - \frac{1}{n}\operatorname{tr} g(D_n) \longrightarrow 0. \tag{5}$$

Proof. Fix $t > 0$. Spectrally truncate $A_n - D_n = R_n + E_n$, retaining in R_n only the eigenvalues whose absolute values exceed t. Then both pieces are Hermitian, $\operatorname{rank} R_n = o(n)$, and $\|E_n\| \leq t$. The min–max principle bounds the difference of the empirical distribution functions of $D_n + R_n$ and D_n by $\operatorname{rank} R_n/n$. For $g \in C_c^1(\mathbb{R})$, integration by parts therefore bounds the difference of their normalized traces by

$$\frac{\operatorname{rank} R_n}{n}\int_{\mathbb{R}} |g'(x)|\,dx.$$

The ordered eigenvalues of $A_n = D_n + R_n + E_n$ and $D_n + R_n$ differ by at most t, again by min–max. Their normalized g-traces differ by at most $\omega_g(t) = \sup_{|x-y|\leq t}|g(x) - g(y)|$. First let $n \to \infty$, then $t \downarrow 0$. Uniform approximation by $C_c^1(\mathbb{R})$ extends the conclusion to all $g \in C_c(\mathbb{R})$. □

Proof of Theorem 1. Take a unitary Schur decomposition

$$V_n^* X_n V_n = L_n + T_n, \qquad L_n = \operatorname{diag}(\lambda_1(X_n), \ldots, \lambda_n(X_n)),$$

where T_n is strictly upper triangular. Put $K_n = \operatorname{Im}(L_n + T_n)$ and $D_n = \operatorname{diag}(\operatorname{Re}\lambda_j(X_n))$. Then $\|K_n\|_1 = \|\operatorname{Im} X_n\|_1 = o(n)$, and a direct entry calculation gives

$$V_n^*(\operatorname{Re} X_n)V_n = D_n + \mathcal{H}(K_n). \tag{6}$$

Indeed $U(K_n) = T_n/(2i)$, so $\mathcal{H}(K_n) = (T_n + T_n^*)/2$. Lemma 1 implies $N_t(\mathcal{H}(K_n)) = o(n)$ for every $t > 0$. Lemma 2 then shows that $\operatorname{Re} X_n$ and D_n have asymptotically equal normalized traces against every continuous compactly supported function on the real line.

The diagonal entries of K_n are $\operatorname{Im}\lambda_j(X_n)$. The diagonal trace-norm bound gives

$$\sum_j |\operatorname{Im}\lambda_j(X_n)| \leq \|K_n\|_1 = o(n). \tag{7}$$

To verify this bound directly, choose a diagonal matrix S whose entries are the signs of these real diagonal entries. Then $\sum_j |(K_n)_{jj}| = \operatorname{tr}(SK_n) \le \|K_n\|_1$, since $\|S\| \le 1$.

Given $F \in C_c(\mathbb{C})$, it is uniformly continuous on $\mathbb{C}$. For every $\delta > 0$, (7) bounds the difference between its eigenvalue averages on X_n and D_n by

$$\omega_F(\delta) + 2\|F\|_\infty \frac{\|K_n\|_1}{n\delta}. \tag{8}$$

Let $n \to \infty$ and then $\delta \downarrow 0$. Thus $D_n \sim_\lambda f$; applying (5) to $F|_{\mathbb{R}}$ and using (6) proves the required conclusion. Finally the limiting probability measure in (1) is supported on $\mathbb{R}$: every nonnegative compactly supported continuous test function away from $\mathbb{R}$ has integral zero. An exhaustion of $\mathbb{C} \setminus \mathbb{R}$ by compact subsets shows that f is real almost everywhere. □

Corollary 1. *If $A_n = A_n^* \sim_\lambda f$ and $\|Y_n\|_1 = o(n)$, then $A_n + Y_n \sim_\lambda f$.*

Proof. Set $X_n = A_n + Y_n$. Since $\|\operatorname{Re} Y_n\|_1, \|\operatorname{Im} Y_n\|_1 \le \|Y_n\|_1$, Lemma 2 gives $\operatorname{Re} X_n \sim_\lambda f$. The Schur argument (6)–(8) compares the eigenvalue averages of X_n and $\operatorname{Re} X_n$ without assuming that either already has a symbol. Their difference tends to zero, proving the claim. □

This also proves the companion perturbation conjecture in [1, 3]. It is recorded here as a corollary of the same argument and receives no separate campaign credit.

Task 2. Determine the sharp triangular counting constant

Find the least dimension-independent C for which $N_t(\mathcal{H}(B)) \le C\|B\|_1/t$ for all Hermitian B. Describe families approaching the optimal constant.

Task 3. Find optimal rates for spectral convergence

Under a uniform density bound for the limiting real eigenvalue law, determine the best rate in Kolmogorov or Lévy distance in terms of $\|\operatorname{Im} X_n\|_1/n$ and the initial spectral approximation error.

Task 4. Classify other spectral supports with this stability

Characterize closed sets $E \subset \mathbb{C}$ for which every normal sequence with spectra in E keeps its spectral law under arbitrary trace-norm $o(n)$ perturbations. Affine real lines satisfy this property; the circular counterexample in [1] shows that it fails for some other supports.

Task 5. Find weaker sufficient hypotheses on imaginary parts

Identify natural conditions on the singular-value tails of $\operatorname{Im} X_n$, weaker than trace-norm $o(n)$, that still guarantee equality of the spectral laws of X_n and $\operatorname{Re} X_n$. Require conditions that also control the triangular transform, since zero singular-value distribution alone is insufficient.

References

[1] G. Barbarino, *Conjectures on Perturbations of Hermitian Sequences*, arXiv:1808.05555v1, Conjectures 1–2. https://arxiv.org/abs/1808.05555.

[2] M. Caspers, D. Potapov, F. Sukochev and D. Zanin, *Weak type estimates for the absolute value mapping*, arXiv:1309.3378; Journal of Operator Theory (2015). https://arxiv.org/abs/1309.3378.

[3] G. Barbarino and S. Serra-Capizzano, *Non-Hermitian perturbations of Hermitian matrix-sequences and applications to the spectral analysis of the numerical approximation of partial differential equations*, Numerical Linear Algebra with Applications (2020), Conjecture 1. https://giovannibarbarino.github.io/doc/articles/NHperturbation.pdf.